

## **Assignment no # 01**

A cybersecurity authentication system processes four interdependent security parameters: user credentials, device verification, network trust score, and session behavior. These parameters are modeled through the following system of linear equations:

$$x + 2y - z + w = 3$$

$$2x + 4y - 2z + 2w = 6$$

$$x + y + x + w = 2$$

$$3x + 6y - 3z + 3w = k$$

Where  $k$  is a real parameter representing a system threshold.

A.

Construct the augmented matrix of the given system and apply row operations to reduce it to echelon form.

B.

Determine the rank of the coefficient matrix and the rank of the augmented matrix. Analyze the relationship between them.

C.

Use the consistency criteria to conclude whether the system is consistent or inconsistent. Interpret your result in the context of system reliability.

D.

Find the value(s) of  $k$  for which the system is:

- consistent (with condition), and
- inconsistent

Respond with proper explanation based on rank conditions.